# Homework Assignment

## Regression and Classification: Error Analysis

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**Submission Deadline: September 10-th**

## Objective

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The purpose of this assignment is to develop a deep understanding of regression and classification models through systematic error analysis. Students are expected to investigate model behavior, identify failure patterns, and distinguish between errors arising from data issues, model assumptions, and problem formulation.

All analyses must be conducted using $k$-fold cross-validation. Students are required to clearly justify their choice of $k$, taking into account factors such as dataset size, computational complexity, and the bias–variance trade-off.

## 1 Regression Error Analysis

### 1.1 Residual Analysis

For each observation, define the residual as:

$$\text{Residual} = y - \hat{y}.$$

**Tasks:**

- Plot residuals as a function of predicted values.
- Plot the distribution of residuals (e.g., histogram).

**Questions**

- Are the residuals centered around zero?
- Do the residuals exhibit any systematic patterns?
- Is there evidence of heteroscedasticity (i.e., non-constant variance)?
- Is there a sector that is more likely to have high error rates? If so, why do you think this is happening?

### 1.2 Error as a Function of Features

**Tasks:**

- Plot feature values versus residuals.
- Plot feature values versus absolute errors.

**Objective:** Identify non-linear relationships, feature-dependent error patterns, and hidden subpopulations.

### 1.3 Analysis of Extreme Errors

**Tasks:**

- Identify the top 5% largest absolute errors.
- Analyze these observations individually.

**Discussion:** Determine whether errors arise from:

- Data quality issues
- Model limitations
- Rare or exceptional cases

### 1.4 Statistical Properties of Errors

**Compute:**

- Mean Absolute Error (MAE)
- Standard deviation of residuals
- Skewness and kurtosis

**Interpretation:**

- Heavy tails may indicate instability
- Skewness may indicate systematic bias

## 2 Regression Models

Students are required to implement and compare multiple regression models. The goal is to understand how different modeling assumptions affect performance and error behavior.

### Required Models

Students must use at least the following three types of models:

- **Linear Regression**
  Serves as a baseline model. Assumes a linear relationship between features and the target variable.
- **Decision Tree Regressor**
  A non-linear model that partitions the feature space into regions. Useful for capturing interactions and non-linear patterns.

- **Ensemble Model (Choose at least one)**

  - Random Forest Regressor
  - Gradient Boosting (e.g., XGBoost, LightGBM, or CatBoost)

  These models typically improve performance by combining multiple learners and reducing variance or bias.

### Optional Models (Bonus)

Students may include additional models for comparison:

- K-Nearest Neighbors (KNN) Regressor
- Support Vector Regression (SVR)

### Requirements

- All models must be trained on the same feature set for fair comparison.
- Hyperparameters should be clearly specified (default or tuned).
- Model performance must be evaluated using multiple metrics.

### Discussion

Students are required to provide a critical analysis of the evaluated regression models.
The discussion should address the following aspects:

- Relative model performance across evaluation metrics (e.g., MSE, RMSE, $R^2$)
- The extent to which model assumptions (e.g., linearity, independence, homoskedasticity) are satisfied
- Differences in the bias–variance trade-off among the models
- Sensitivity of the models to outliers and noise
- Identification of scenarios in which linear models fail, and an explanation of why
- Situations in which tree-based models provide advantages
- The trade-off between interpretability and predictive performance
- Selection of a preferred model for future analysis, supported by a clear and well-reasoned justification
- What did you learn from the modeling stage?

# 3 Classification Error Analysis

## 3.1 Confusion Matrix Analysis

**Tasks**

- Present, visualize, and provide a detailed interpretation of the confusion matrix
- Analyze the False Positives (FP) and False Negatives (FN), and discuss their implications

**Questions:**

- Which type of error is most critical in this context?
- In which regions does the model fail systematically?
- What insights can be derived from the analysis?

## 3.2 Probability-Based Analysis

**Tasks:**

- Compare predicted probabilities for correct vs incorrect predictions

**Objective:** Identify high-confidence errors.

## 3.3 Error as a Function of Features

**Tasks:**

- Compare feature distributions for correct vs incorrect predictions

**Objective**

Identify regions prone to misclassification.

This analysis is important for understanding the limitations of the model, detecting systematic failure patterns, and identifying subpopulations where performance degrades. Such insights can guide feature engineering, model selection, and the design of more robust and reliable predictive systems.

## 3.4 Threshold Sensitivity Analysis

Evaluate performance across thresholds:

$$0.1, 0.2, \ldots, 0.9.$$

**Compute:**

- Precision
- Recall
- $F_1$-score
- Plot the $F_\beta$-score as a function of $\beta$
- Matthews Correlation Coefficient (MCC)
- Area Under the ROC Curve (AUC–ROC), including the ROC curve

**Analysis:**

- Trade-off between false positives and false negatives
- Identify stable and unstable operating regions

### 3.5 Discussion

Students are required to provide a critical comparison of the evaluated models.

Students are required to provide a critical reflection on the overall analysis, integrating insights from both the modeling and error analysis stages.

The reflection should address:

- The strengths and limitations of the evaluated models in relation to the data.
- Key failure modes identified during the analysis.
- The relationship between model assumptions and observed performance.
- The most significant insights derived from the study.
- Concrete recommendations for improving model performance and robustness in future work.

## 4 Final Reflection

Provide a concise discussion addressing:

1. Where does the model fail most?
2. Are failures due to data, model, or formulation?
3. What improvements would you propose?
4. What insights did you gain in your analysis?